

Module 4

The Time Value of Money

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Goals

- **Part 1**
 - Translate dollars today into dollars tomorrow (“future value”)
 - Translate dollars tomorrow into dollars today (“present value”)

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- **Part 3**
 - Special cases, perpetuities, annuities, and their variations

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 - Translate dollars today into dollars tomorrow (“future value”)
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- **Part 2**
 - Financial calculators
- **Part 3**
 - Special cases, perpetuities, annuities, and their variations
- **Part 4**
 - The role of inflation

Module 4, Part 1

Future Value and Present Value

The Time Value of Money



Would you rather receive a dollar today, or a dollar tomorrow?

The Time Value of Money

- A dollar today is worth more than a dollar tomorrow

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- Why?

The Time Value of Money

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- Why?
- Because we can invest that dollar today and collect interest on our investment!

The Time Value of Money

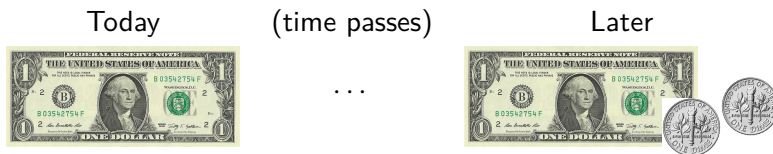
- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!
- Assuming,
 - There is some positive rate of return investment for us to pursue
 - No behavioral concerns that make us poor with money

Future Values

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- A “present value” today grows to its “future value” tomorrow via interest

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$$\text{Future value (FV)} = \$100 \times (1 + 0.06)^5 = \$133.82$$

Compound Interest

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- Compound interest is the default in finance (e.g., mortgage rate, deposit rate)

Future Value, Compound Interest

- You have \$100 in a bank account, and your bank pays 6% per year in interest
- How much interest will you have earned after five years?

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Year	Start balance	Compound interest	End balance
1	\$100.00	$0.06 \times 100.00 = \$6.00$	\$106.00
2	\$106.00	$0.06 \times 106.00 = \$6.36$	\$112.36
3	\$112.36	$0.06 \times 112.36 = \$6.74$	\$119.10
4	\$119.10	$0.06 \times 119.10 = \$7.15$	\$126.25
5	\$126.25	$0.06 \times 126.25 = \$7.57$	\$133.82

Future Value, Compound Interest

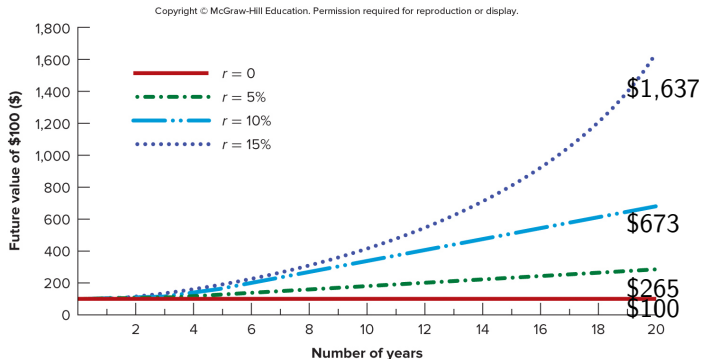
- You have \$100 in a bank account, and your bank pays 6% per year in interest
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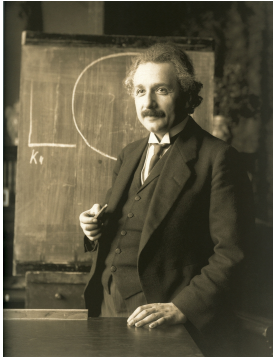
- With compound interest, your \$100 grows to $\$100(1 + r)^T = 133.82$, so you earned \$33.82 in compound interest

Future Value

- The future value (FV) depends heavily on the interest rate r , especially as the time dimension t grows

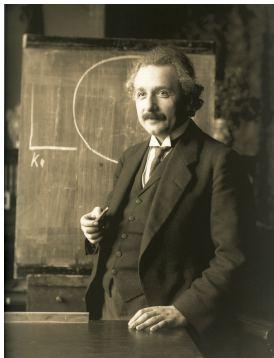


What's the Most Powerful Force in the Universe?



- Einstein, ...

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- Einstein, ...
- “Compound interest”

Returns, Holding Period and Annualized

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- These financial returns are not directly related in any way with **Module 3's** accounting-based ROA and ROE measures

Example, Annualized Returns

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- Notice that the annual return is NOT $28.2\%/2 = 14.1\%$

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- Notice that the annual return is NOT $28.2\%/2 = 14.1\%$
- Confirm, you can also verify that $(1 + 0.132)^2 - 1 = 28.2\%$

Example, Future Value

- Peter Minuit bought Manhattan Island for \$24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
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- b. $FV = \$24 \times (1 + 0.035)^{395} = \19 million



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- 11.896 years, because $(1 + 0.06)^{11.896} = 2$



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- **Discount rate**, the rate used to compute PVs (i.e., r)
 - For a guaranteed future cash flow, the correct discount rate is the risk-free interest rate
 - For a (systematically) risky future cash flow, we may need a higher discount rate to account for the higher risk, we will return to this in **Modules 8 and 9**

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- **Discount factor**, the present value of \$1 received in year t when the discount rate is r , less than one when r is positive
- **Discounted cash flow (DCF)**, the method of calculating present values by discounting future cash flows

Example, Present Value

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

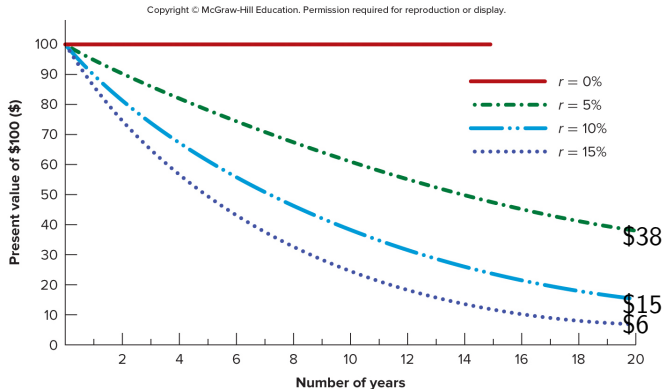
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$$\text{Present value (PV)} = \frac{\text{FV}}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

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Present Value, Multiple Future Cash Flows

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$$PV_{\text{total}} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots$$

- where C_1 is the cash flow in one year, C_2 is the cash flow in two years, etc.

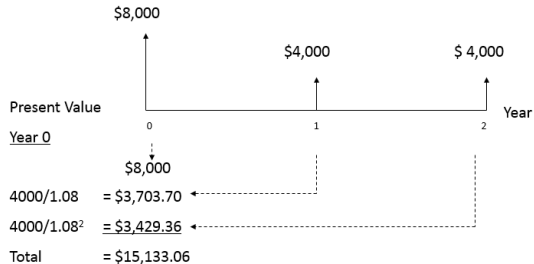
Example, Multiple Future Cash Flows

- Your auto dealer gives you the choice to either
 - A. Pay \$15,500 cash now, or
 - B. Make three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

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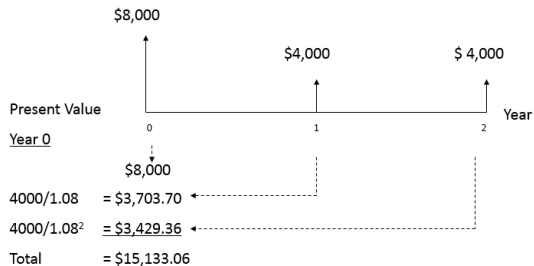
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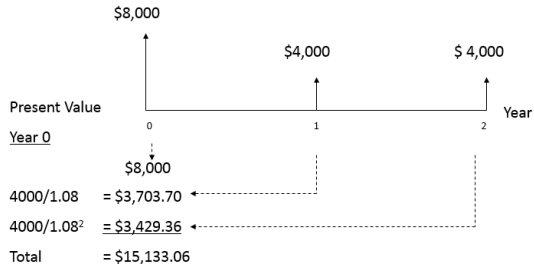
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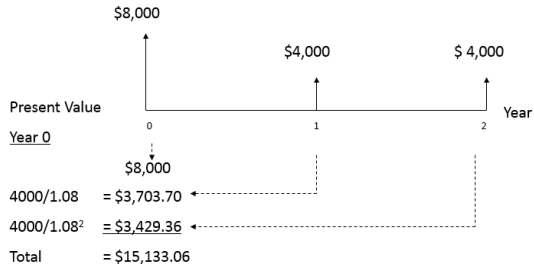


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- The present value (\$15,133.06) is less than \$15,500, so choose the payment plan
- This is the principle behind the NPV rule in capital budgeting, which we will return to in **Modules 6 and 7**

Recap of Part 1

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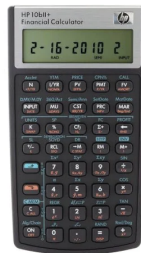
- Present values can be added together to calculate the total present value of multiple future cash flows

Module 4, Part 2

Financial Calculators

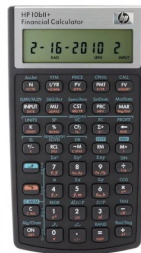
Financial Calculators

- Speeds up time value of money problems (and solves harder ones!)



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- Speeds up time value of money problems (and solves harder ones!)
- A complement to the underlying math and intuition, never a substitute



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- After you reset, you should repeat the next two settings!

Set the Number of Significant Digits to at Least 4!

- On the HP 10bII+, to change the number of digits displayed to 4 decimals, press
 - OrangeOption
 - = (DISP)
 - 4

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 - .
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 - 1
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- On the HP 10bII+, to set it to 1, press
 - 1
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- To change back to the factory default (12), press
 - 12
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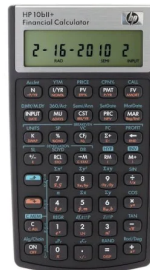
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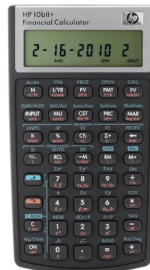
Financial Calculators, Basic Functionality

- N, number of periods (usually years)



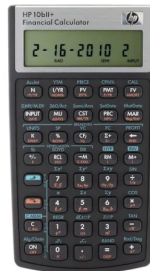
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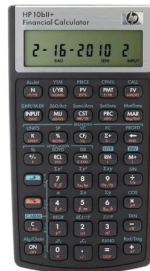
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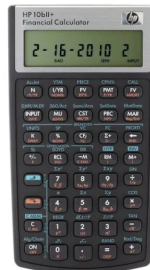
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- FV, future value (lump sum)



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n	i	PV	PMT	FV
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Example, Puerto Rico Borrows

- In 2007, Puerto Rico needed to borrow \$2.6 billion for 47 years
- It did so by selling bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each (they sold around $\$2.6 \text{ billion} / 94.40 = 27.5 \text{ million}$ bonds, and thus promised to pay back around \$27.5 billion in total)
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- Bonus, what would the payment be if the monthly interest was 0.2%?

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- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?

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- Let's try to solve some of the previous examples using Excel!

Example, A-Rod's \$275 Million Contract with the Yankees

Year	Salary	"Signing bonus"
2008	27	2
2009	32	1
2010	32	1
2011	31	1
2012	29	1
2013	28	1
2014	25	3
2015	21	
2016	20	
2017	20	
Total	265	10

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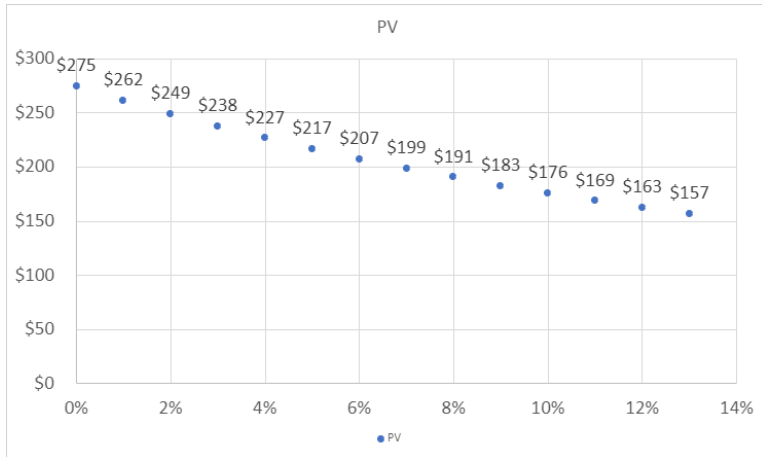
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- What was A-Rod's \$275 million contract with the Yankees actually worth?
- Suppose the discount rate is 10%. What is the present value of his contract?
- Why might it be useful for A-Rod to know the PV, and not simply the (undiscounted) sum of all future payments?



The PV of Payments to A-Rod



Recap of Part 2

- Financial calculators solve any one of N , I/YR , PV , PMT , and FV from the other four

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- The PV has the opposite sign of the payments (PMT) and the future value (FV)
- Set the number of compounding periods per year to 1, and show at least 4 significant digits
- Excel has the same functionality, with the rate in decimals

Module 4, Part 3

Special Cases: Perpetuities and Annuities

Perpetuities and Annuities

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- Why care about perpetuities and annuities?
 - Many real-world cash flows have these patterns (particularly annuities, e.g., most mortgages or other fixed-term loans)
 - There exist neat shortcuts that make it very easy to calculate the present value of these patterns of cash flows

Perpetuity Structure

- A perpetuity is a stream of equal cash flows (C) that occur at regular intervals and last forever



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- C is the annual cash flow (aka the payment), and r is the discount rate

Example, Perpetuities

- Recall A-Rod's Yankees contract. Suppose the discount rate is 10%, then the PV of the contract (paying \$275m over 10 years) is \$175.6m
- Now suppose the Yankees offer A-Rod two options,
 - A. The contract as previously described (10 years with \$275m in total), or
 - B. A perpetual payment of C dollars every year. He will play for 10 years, but continue to be paid forever, as long as he lives, and then his children, and then his children's children, etc.
- How much should the Yankees offer as the perpetual payment C for him to be **indifferent** between contract (A) and the “being paid forever” contract (B)?



Example, Perpetuities

- Solution, he is indifferent if the contracts have the same present value!

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$$PV_{\text{Contract A}} = PV_{\text{Contract B}} = \frac{C_B}{r} \Rightarrow C_B = PV \times r = 175.6 \times 10\% = \$17.56m$$

Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

$$PV = \frac{C}{r} \times (1 + r)$$



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Perpetuity, Timing

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- **Ordinary perpetuity**, first payment at time 1

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- **Delayed perpetuity**, first payment at time $t > 1$

$$PV = \frac{C}{r} \times \frac{1}{(1+r)^{t-1}}$$



Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at C but grows at a constant growth rate g forever. This is a growing perpetuity

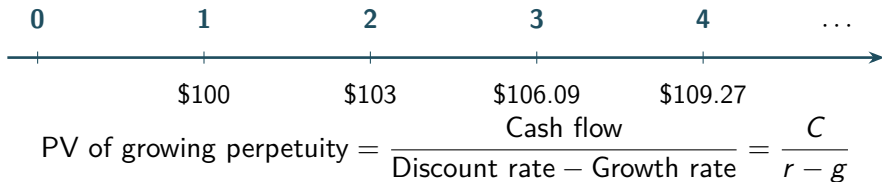
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- Example, a growing perpetuity with a first payment of \$100 that grows at a rate of 3% has cash flows



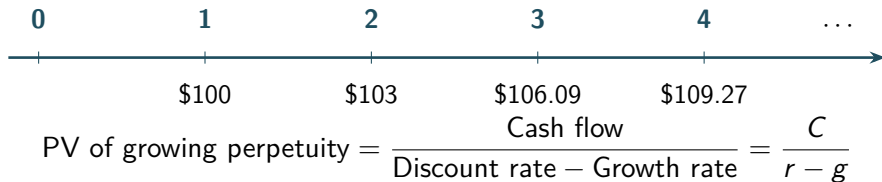
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- This formula comes up often when valuing stocks, e.g., in the “Gordon growth model” and DCF valuation (more in [Modules 5 and 9](#))
- When $g = 0$, the PV simplifies to the formula for a regular (non-growing) perpetuity

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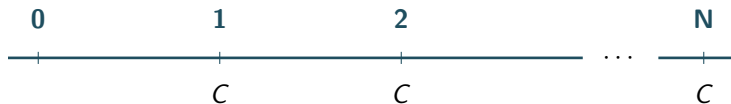


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- Present value of an annuity,

$$\text{PV of annuity} = C \underbrace{\frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right)}_{\text{present value annuity factor (PVAF)}}$$

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- C is the annual cash flow (aka the payment), r is the discount rate, and N or T is the number of years that the annuity lasts

Example, Annuity

- You are purchasing a car. You are scheduled to make 3 annual installments of \$8,000 per year, with the first payment a year from now. Given a discount rate of 10%, what is the price you are paying for the car (i.e., what is the PV of the annuity)?

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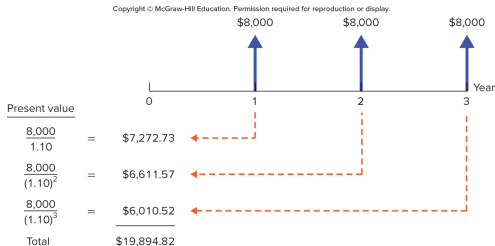
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$$PV = C \frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right) = \$8,000 \times \frac{1}{0.10} \times \left(1 - \frac{1}{(1+0.10)^3} \right) = \$19,895$$

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Example, Annuity

- Suppose the discount rate for A-Rod's contract is 10%, so the present value is \$175.6m
- Now suppose instead the Yankees offer an annuity for 10 years with the same present value
- How much would he get paid per year, and in total (undiscounted) over 10 years, under such a same-value annuity contract?



Example, Annuity

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$$175.6 = C \frac{1}{10\%} \left(1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

Example, Annuity

- Solution,

$$175.6 = C \frac{1}{10\%} \left(1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

- So he would be paid \$28.58m per year, or \$285.8m in undiscounted total over 10 years (slightly more in total, because his actual contract is front-loaded, which raises its present value compared to a constant payment)

Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?

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Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?
- Answer, the front-loaded one
- What about a back-loaded cash flow and an annuity?

Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

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- **Ordinary annuity**, first payment at time 1

$$PV = PV \text{ annuity}$$



Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



- **Ordinary annuity**, first payment at time 1

$$PV = PV \text{ annuity}$$



- **Delayed annuity**, first payment at time $t > 1$

$$PV = PV \text{ annuity} \times \frac{1}{(1+r)^{t-1}}$$



Practice Problem

- A store offers two payment plans. Under the installment plan, you pay 25% down and 25% of the purchase price in each of the next 3 years. If you pay the entire bill immediately, you can get a discount of 7% on the purchase price. Assume the product sells for \$100
- a-1. Calculate the present value of the payments if you can borrow or lend funds at an interest rate of 4%
- a-2. Which is a better deal?
- b-1. Calculate the present value if the payments on the 4-year installment plan do not start for a full year
- b-2. Which is a better deal?

Practice Problem

- a-1. This is an annuity due, since the first payment is due today

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$$PV = \$25 + [\$25 \times \text{annuity factor (4\%, 3 years)}]$$

$$PV = \$25 + \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^3} \right] = \$94.38$$

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- a-2. Paying in full now is better (\$93 beats \$94.38)
- On a financial calculator, $PMT = -25$, $N = 3$, $FV = 0$, $I/YR = 4$, leading to 69.38, and next, $69.38 + 25 = 94.38$

Practice Problem

- b-1. Now it is an ordinary annuity, since the first payment is due at the end of year 1

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Practice Problem

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$$PV = \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^4} \right] = \$90.75$$

- b-2. The installment plan is better (\$90.75 beats \$93)
- On a financial calculator, $PMT = -25$, $N = 4$, $FV = 0$, $I/YR = 4$

Practice Problem

- A local bank advertises the following deal, “Pay us \$100 at the end of each year for 11 years, and then we will pay you (or your beneficiaries) \$100 at the end of each year forever”
 - a. Calculate the present value of your payments to the bank if the interest rate is 7.00%
 - b. What is the present value of a \$100 perpetuity deferred for 11 years if the interest rate is 7.00%?
 - c. Is this a good deal?

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

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$$PV = C \left(\frac{1}{r} \right) \left(1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left(\frac{1}{7\%} \right) \left(1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

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- On a calculator, PMT = 100, I/YR = 7, N = 11, FV = 0. Find PV

Practice Problem

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- b. The bank's payments to you are a perpetuity deferred for 11 years

Practice Problem

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- On a calculator, PMT = 100, I/YR = 7, N = 11, FV = 0. Find PV
- b. The bank's payments to you are a perpetuity deferred for 11 years

$$PV_{11} = \frac{C}{r}$$
$$PV_{11} = \frac{100}{7\%} = \$1,428.57$$
$$PV = \frac{PV_{11}}{(1+r)^{11}} = \frac{1,428.57}{(1+7\%)^{11}} = \$678.70$$

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- c. Not a good deal (you pay \$749.83 for something worth \$678.70)

Growing Annuities

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$$\text{PV of growing annuity} = \frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right]$$

Take the Lump Sum or Growing Annuity?

- The Robinsons of Munford, Tennessee were one of the three winners in the “1.6 billion” Powerball jackpot in 2016
- They could choose between,
 - A. A lump sum of \$327 million, or
 - B. 30 installments of a growing annuity due with $g = 4\%$, starting at \$9.43m, with 30 payments, totaling around \$529 million



Photo, AP/Mark Humphrey

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- Which option should they take?



Photo, AP/Mark Humphrey

Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ($g = 4\%$), it starts immediately and grows for another 29 years at 4% and then ends

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ($g = 4\%$), it starts immediately and grows for another 29 years at 4% and then ends
- Suppose the discount rate is 3%

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$

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0	9.43
1	9.81
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Take the Lump Sum or Growing Annuity?

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$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$
$$= \frac{9.43}{0.03 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$

Year	Payout
0	9.43
1	9.81
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Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\begin{aligned} & \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^N \right] \times (1+r) \\ &= \frac{9.43}{0.03-0.04} \left[1 - \left(\frac{1+0.04}{1+0.03} \right)^{30} \right] \times (1+0.03) \\ &= \$326.58\text{m} \end{aligned}$$

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
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29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

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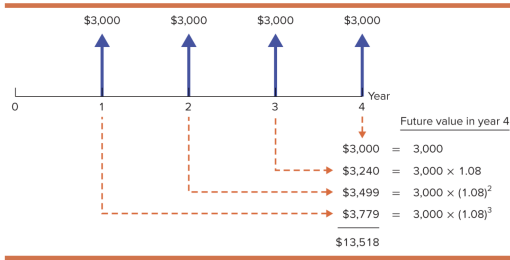
$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$
$$= \frac{9.43}{0.03 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$
$$= \$326.58\text{m}$$

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
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Total	528.88

- That is, the present value of the annuity is almost exactly the same as (but slightly lower than) the \$327m lump sum

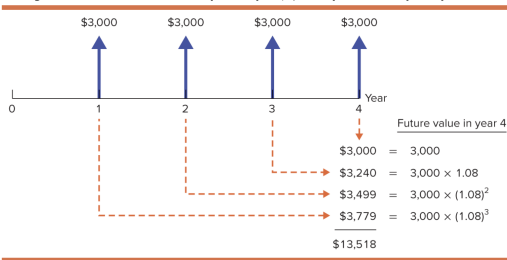
Future Value of Annuities

FIGURE 5.11 Calculating the future value of an ordinary annuity of \$3,000 a year for four years (interest rate = 8%).



Future Value of Annuities

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FV of annuity of \$1 a year = PV of annuity of \$1 a year $\times (1 + r)^t$

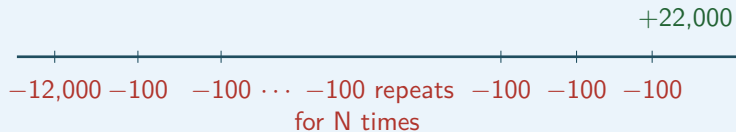
$$= \left[\frac{1}{r} - \frac{1}{r(1 + r)^t} \right] \times (1 + r)^t = \frac{(1 + r)^t - 1}{r}$$

Practice Problem

- I now have \$12,000 in the bank earning interest of 0.50% per month. I need \$22,000 to make a down payment on a house. I can save an additional \$100 per month. How long will it take me to accumulate the \$22,000?

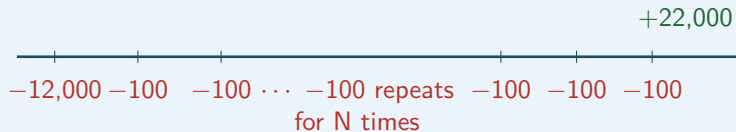
Practice Problem

- From your perspective, the cash flow is



Practice Problem

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- You need to solve, future value (FV) of the \$12,000 + FV of the N \$100s = FV of the \$22,000

$$12000 \cdot (1 + 0.005)^N + 100 \cdot \frac{(1 + 0.005)^N - 1}{0.005} = 22000$$

Practice Problem

- The problem can, alternatively, be solved as, present value (PV) of the \$12,000 + PV of the N \$100s = PV of the \$22,000

Enter		0.50	-12,000	-100	22,000
	N	I/Y	PV	PMT	FV
Solve for	54.52				

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	N	I/Y	PV	PMT	FV
Solve for	54.52				

- Let's recall that financial calculators are programmed to solve the following,

$$PV + \left[\frac{PMT}{(1 + I)^1} + \frac{PMT}{(1 + I)^2} \cdots + \frac{PMT}{(1 + I)^N} \right] + \frac{FV}{(1 + I)^N} = 0$$

Practice Problem

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Enter		0.50	-12,000	-100	22,000
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Enter		0.50	-12,000	-100	22,000
	N	I/Y	PV	PMT	FV
Solve for	54.52				

Practice Problem

- A couple will retire in 50 years. They plan to spend about \$20,000 a year in retirement, which should last about 25 years. They believe that they can earn a real interest rate of 9% on retirement savings
 - a. If they make annual payments into a savings plan, how much will they need to save each year? Assume the first payment comes in 1 year
 - b. How would the answer to part (a) change if the couple also realize that in 20 years they will need to spend \$50,000 on their child's college education?

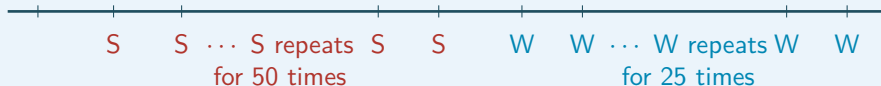
Practice Problem

- a. The cash flow is

Time 0, today

Time of retirement

Final withdrawal



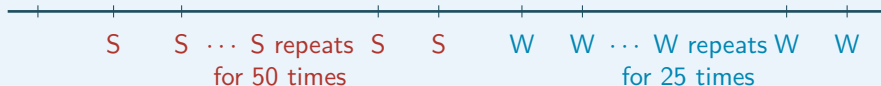
Practice Problem

- a. The cash flow is

Time 0, today

Time of retirement

Final withdrawal



- How much savings is required at the time of retirement?

$$PV_{\text{withdrawals, year 50}} = 20,000 \frac{1}{0.09} \left(1 - \frac{1}{(1 + 0.09)^{25}} \right) = 196,451.59$$

- On a calculator, $PMT = 20,000$, $I/YR = 9$, $N = 25$, $FV = 0$

Practice Problem

- The FV of the 50 savings, each of the amount $\$S$, should be that

Practice Problem

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$$FV_{\text{savings, year 50}} = S \left(\frac{(1 + 0.09)^{50} - 1}{0.09} \right) = 196,451.59 \Rightarrow S = \$241.02$$

- On a calculator, $FV = 196,451.59$, $I/YR = 9$, $N = 50$, $PV = 0$

Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and, $PV \text{ of all savings, discounted to today} = PV \text{ of all withdrawals, discounted to today}$

Practice Problem

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- The first step is to determine the PV of the cash needed for both college and retirement at time 0

$$PV = \frac{50,000}{1.09^{20}} + \frac{196,451.59}{1.09^{50}} = 11,563.53$$

Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and, PV of all savings, discounted to today = PV of all withdrawals, discounted to today
- The first step is to determine the PV of the cash needed for both college and retirement at time 0

$$PV = \frac{50,000}{1.09^{20}} + \frac{196,451.59}{1.09^{50}} = 11,563.53$$

- Now determine the annual savings for 50 years that equates to that present value

$$\$11,563.53 = C \times \frac{1 - \frac{1}{(1+9\%)^{50}}}{9\%} \Rightarrow C = \$1,054.90$$

- On a calculator, PV = 11,563.53, I/YR = 9, N = 50, FV = 0

Practice Problem

- Professor's Annuity Corporation offers a lifetime annuity to retiring professors. For a payment of \$79,000 at age 65, the firm will pay the retiring professor \$575 a month until his death
- a. If the professor's remaining life expectancy is 20 years, what is the monthly interest rate on this annuity?
- b. If the monthly interest rate is 1.00%, what monthly annuity payment can the firm offer to the retiring professor?

Practice Problem

- a. The monthly payments are a 240-month ordinary annuity

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$$\$575 \times \left[\frac{1}{r} - \frac{1}{r \times (1+r)^{240}} \right] = \$79,000 \Rightarrow r = 0.52\%$$

- On a calculator, $PMT = 575$, $N = 240$, $PV = -79,000$, $FV = 0$. Find I/YR

Practice Problem

- a. The monthly payments are a 240-month ordinary annuity

$$\$575 \times \left[\frac{1}{r} - \frac{1}{r \times (1+r)^{240}} \right] = \$79,000 \Rightarrow r = 0.52\%$$

- On a calculator, $\text{PMT} = 575$, $N = 240$, $\text{PV} = -79,000$, $\text{FV} = 0$. Find I/YR
- b.

$$C \times \left[\frac{1}{0.01} - \frac{1}{0.01 \times (1.01)^{240}} \right] = \$79,000 \Rightarrow C = \text{PMT} = \$869.86$$

- On a calculator, $N = 240$, $\text{PV} = -79,000$, $\text{FV} = 0$, $\text{I/YR} = 1$. Find PMT

Recap of Part 3

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Recap of Part 3

- A perpetuity is a level stream of cash flows at regular intervals that never ends
- An annuity is a level stream of cash flows at regular intervals that ends after N periods (usually years, but the periods do not have to be years)
- These have handy formulas for calculating their present values!
- Timing,
 - Ordinary perpetuities and annuities make their first payment at time $t = 1$
 - Perpetuity and annuity dues make their first payment at $t = 0$
 - Delayed perpetuities and annuities make their first payment at $t > 1$

Module 4, Part 4

Inflation

History of Inflation



Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.
Source: U.S. Bureau of Labor Statistics.

Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)

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 - Compare dollars at time $t = 0$ to dollars at time $t = 1$

Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)
- **Nominal interest or nominal return**, the rate at which money invested grows
 - Compare dollars at time $t = 0$ to dollars at time $t = 1$
- **Real interest or real return**, the rate at which purchasing power increases
 - Compare the ability to buy a basket of goods at time $t = 0$ to the ability to buy the same basket of goods at time $t = 1$

Inflation and Returns

- In the presence of inflation, an investor's real return (e.g., the real interest rate) is always less than the nominal return

$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

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Approximation, $\text{Real return} \approx \text{Nominal return} - \text{Inflation rate}$

- Inflation also affects discounting,
 - We should discount nominal dollar cash flows using a nominal discount rate
 - We should discount real dollar cash flows using a real discount rate

Example, Inflation

- Question, if the interest rate on one-year government bonds is 6.0% and the inflation rate is 2.0%, what is the real interest rate?

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- Answer,

$$1 + \text{Real interest rate} = \frac{1 + \text{Nominal interest}}{1 + \text{Inflation}} = \frac{1.06}{1.02} = 1.039$$

$$\text{Real interest rate} = 0.039 \text{ or } 3.9\%$$

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$$\text{Real interest rate} = 0.039 \text{ or } 3.9\%$$

- Approximation, $\text{real} \approx 6\% - 2\% = 4.0\%$

Key Takeaways

- A dollar today is worth more than a dollar tomorrow, $FV = PV \times (1 + r)^t$ and $PV = FV / (1 + r)^t$ translate between them

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Key Takeaways

- A dollar today is worth more than a dollar tomorrow, $FV = PV \times (1 + r)^t$ and $PV = FV / (1 + r)^t$ translate between them
- Perpetuities and annuities have shortcut PV formulas, mind the timing of the first payment
- The real return strips inflation out of the nominal return

Thank you!

Please let me know if you have any questions.